AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

M269
Algorithms, Data Structures and Computability

Final Examination
Fall Semester 2024/2025

Date: 8 /Jan./2025

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 3 Hours

Instructions:

- Total Marks: **100**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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PART 1: ALL QUESTIONS ARE REQUIRED [10 Marks]

Question 1: (10 marks)

Mark the following statements are true or false

1) Given 32 sorted items in an array, using binary search, the average number of searches that will be required to find an item in such array is 5.
2) Minimal spanning tree algorithm belongs to the brute-force paradigm.
3) The sequence [20, 15, 18, 7, 9, 5, 12, 3, 6, 2] is a max-heap.
4) To represent hierarchical relationship between family members, a graph data structure is suitable.
5) The worst-case performance of looking up an element in a hash table is O(n), where n is the number of elements.
6) Linear search in an array A[n] involves (log n) comparisons.
7) O(n log n), O(n), $O(n^2 \log n)$, $O(n^2)$ is the right order from best to worst complexity.
8) A graph is a collection of nodes, called vertices and line segments called edges that connect pair of nodes.
9) In the breadth first traversal, we process all of a vertex's descendants before we move to an adjacent vertex.
10) KMP searching algorithm has the same speed of regular naïve search.

PART 2: ALL QUESTIONS ARE REQUIRED [40 Marks]

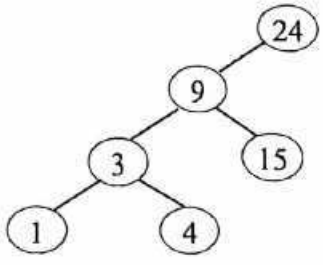
Question 2: (15 marks)

a) Complete the following sentences: (6 marks: 1 mark each)

1) In the ..... paradigm of algorithms, the smallest sub-problems are solved first and results are stored to be used to construct larger sub-problems solution.
2) In the ..... paradigm of algorithms, the solution is constructed through a sequence of steps. At each step, the next optimal step is selected locally, without considering whether this step leads to the final global optimal solution or not.
3) In the ..... search of the graph, we process all of a vertex's descendants before we move to an adjacent vertex.
4) In the ..... problems, computers can quickly verify solutions to these problems, but finding the solutions can be time-consuming.
5) ..... logic is a formal language in which propositions are expressed in terms of predicates, variables and quantifiers
6) ..... is a mathematical model of computation that can be in exactly one of a finite number of states at any given time

b) Consider the following binary search tree (BST). (6 marks: 3 mark each)

1) Draw the same BST after the insertion of keys: 6, 45, 32, 98, 55, and 69, in this order.
2) Draw the BST resulting from the deletion of keys 9 and 45 from the BST resulting from the previous question to produce a balanced tree (i.e., a minimum-height tree).



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c) For the following pattern, generate the failure table to be used in KMP algorithm and state the complexity for both generating such table and the KMP algorithm. (3 marks)

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Question 3: (15 marks)

a) What is the complexity of the following tasks? (10 marks: 1 mark each)

1) Traversing a graph using Breadth First Search algorithm with N nodes and M edges.
2) Finding the largest number of a sorted doubly linked list.
3) Computing the mathematical average of all items in a queue of size 2n.
4) Searching for an element using hash tables with chaining.
5) Finding the third smallest element in an unordered list using the Quick select algorithm.
6) Searching for a pattern of size M into a string of size N using Naïve algorithm.
7) Building the max heap tree from an unordered list.
8) Calling two sorting algorithm consecutively to sort the same list of n randomly arranged numbers: first call Heap-Sort and then its output is processed by Bubble-Sort.
9) Finding the set of ancestors of a node in a balanced binary tree.
10) The brute force approach of finding the common elements among three lists of length N.

b) Demonstrate the insertion of keys 5, 28, 19, 15, 20, 33, 12, 17, 10 into a hash table with collisions resolved by open addressing. Let the hash table have 9 slots, and let the hash function be h(k) = k mod 9. (5 marks)

Question 4: (10 marks)

a) The following famous algorithms belong to different paradigms of algorithms. For each algorithm, state the name of the paradigm and briefly describe it in your won words (total 4 marks: 2 marks each)
   1) Merge sort algorithm
   2) Bubble sort algorithm

b) Give a brief answer to the following questions: (4 marks)
   1) Given an array A[20,60,80,90,90,100,110] and key = 90; In how many iterations would the linear search and binary search reach to the key? (2 marks)
   2) What is the name of the algorithm that is able to find the kth smallest element in an unordered list (1 mark)?
   3) If several elements are competing for the same bucket in the hash table, what is it called? list two techniques to avoid this scenario? (3 marks)

PART 3: ALL QUESTIONS ARE REQUIRED [50 Marks]

Question 5: (40 marks)

Consider the weighted graph illustrated in Fig.1

a) Does this graph contain Euler circuit or/and Euler path? Why? (4 marks)

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b) What is the sequence of visited node in BFS, and DFS if both algorithms start from node A? (4 marks)

c) Draw the minimum spanning tree of this graph. (6 marks). What is the name and complexity of such algorithm? (4 mark)

d) Use the Dijkstra's algorithm to determine the shortest paths from node (A) to all other cities (8 marks). Determine the shortest path and cost from node A to node H (2 marks). [Hint: use the Dijkstra's table to implement the algorithm step by step to show which node will be added in sequence]

e) Implement the proper algorithm to find the shortest paths between all pairs of nodes [Compute D_0, D_1, D_2]. Mention the name of the algorithm that could be used and its complexity (12 marks)

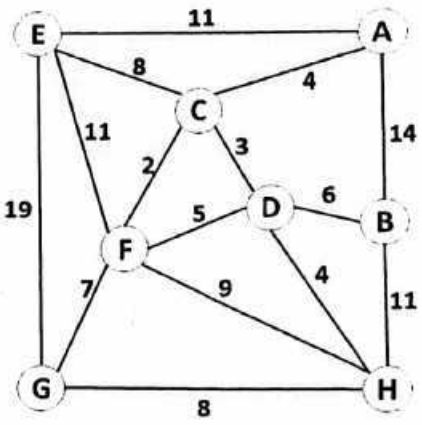


Figure 1 weighted graph

Question 6: (10 Marks)

Playing with regular domino cards with identical heights. The domino cards are arranged in a line, each centered at a specific coordinate on the X-axis. When a domino is pushed to the right, it may cause other dominoes to fall if it touches the next one in the line (i.e., if the height of the card is bigger than the distance between the two consecutive cards)

Input:

- N = number of dominoes
- Height = The height of any domino's card:
- Position[ ] = array of the x-coordinates of the n domino cards.

Output:

- Print *n* space-separated integers, where each integer represents the number of dominoes that would fall if the corresponding domino is pushed to the right.

	Input	output	Explain
Test case 1:	N=4 Height= 5 Position[16,20,10,18]	3 1 1 2	It means that if we push to the right the domino in x-coordination 16, 3 dominos will fall including itself, and so on.
Test case 2:	N=6 Height= 3 Position[5,10,9,1,12,3]	1 2 3 3 1 2	It means that if we push to the right the domino in x-coordination 5, only I card will fall (itself), and so on.

Answer the following questions:

a) Design an algorithm to solve this problem (6 marks)

b) Derive the complexity of your algorithm. (4 marks)

End of questions

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